

Individual Addendum Template
QM 2023 Capstone Project - Milestone 4
Individual Contribution Statement
Name: Nevaeh Marquez
Team: First Row Team
Date: 5/1/26

1. Personal Contribution to Capstone Milestones

Milestone 1: Data Pipeline (Week 5)

Tasks completed:

- Imported and organized all datasets used for the project (housing, hurricane, and hurricane costs)
- Identified and selected the final variables/data sources the team would use
- Collaborated with team members to finalize the research question

Hours spent: 6-8 hours

Key deliverable: Initial cleaned data and finalized research question

Milestone 2: EDA Dashboard (Week 10)

Tasks completed:

- Created multiple visualizations (scatter plots, relationships between housing and hurricane data, etc.)
- Helped analyze trends and patterns from the visuals
- Completed the Data Quality Report

Hours spent: 5-6 hours

Key deliverable: Visualizations and Data Quality Report for M2

Milestone 3: Econometric Models (Week 14)

Tasks completed:

- Completed the Data Quality Report updates
- Updated the README file to clearly explain project structure and workflow
- Helped write and refine the AI Audit documentation

Hours spent: 4-5 hours

Key deliverable: Update README and AI Audit

Milestone 4: Final Investment Memo (Week 14)

Tasks completed:

- Created and designed the final presentation slides

- Helped ensure the presentation communicated findings in a simple, understandable way so the class could understand

Hours spent: 3-4 hours

Key deliverable: Final presentation slides

Total Estimated Contribution

Total hours across all milestones: 25 hours

Percentage of team workload: 25%

Role(s) on team: Data preparation support (M1), Visualization and data quality lead (M2),

Documentation support (M3), Presentation lead (M4)

2. One Defended Methodological Decision

Decision: I supported focusing on visualizing the relationship between hurricane activity and housing prices before building models.

Reasoning: In M2, our visualizations showed clear patterns between hurricane damage and increases in housing prices and rent, which helped us decide what variables to use later.

Economically, this makes sense because when hurricanes damage homes, people need to rebuild, which increases demand and can push prices higher. Starting with visuals also made our approach stronger since we based our decisions on real patterns in the data instead of just guessing.

Alternative considered: We could have jumped straight into modeling, but that would have been less effective because we wouldn't fully understand the data patterns first.

3. One Key Limitation of Our Analysis

Limitation: Our analysis may not fully capture all factors that affect housing prices beyond hurricanes.

Why this matters: Housing prices are influenced by many things like interest rates, population growth, and economic conditions. If those factors change at the same time as hurricanes, it can make it harder to isolate the true impact of hurricanes on housing prices.

Potential mitigation: Future research could include more control variables (like interest rates or income levels) or use a more advanced model to better isolate hurricane effects.

4. AI Audit Notes (If Applicable)

AI Tools Used: GitHub Copilot

Specific AI Use Examples:

Example 1: Importing Data

Prompt: "Help me import this data" (with the dataset URL pasted in the prompt)

Output: AI generated code to load the provided data sources into data frames/files for use in the project workflow.

Verification: We checked that the imported data matched the source data (file contents/structure looked correct), then shared results with Dr. Seagraves for confirmation.

Critique: [What the AI got wrong, how you corrected it]

Example 2: Data Quality Report

Prompt: Help me format the data quality report for M2

Output:

Verification:

Critique: [What the AI got wrong, how you corrected it]

Overall AI Use: I used AI for about 20–30% of my work, mainly for writing support and organization. All final decisions and interpretations were done by me.

5. Self-Reflection

What did I do particularly well on this capstone?

I did a really good job creating visuals and making the presentation clear and easy to understand. I also helped keep our work organized through documentation.

What could I have improved?

I could have started earlier on some parts, especially documentation updates, so I wasn't doing them close to deadlines.

What did I learn from this capstone project?

I learned how important data cleaning and visualization are before doing analysis. I also learned how to work as a team on a long project and communicate ideas clearly, especially when turning data into something people can actually understand.

6. Attestation

By submitting this individual addendum, I affirm that:

All contributions listed above are accurate and honest

I have not exaggerated my role or minimized teammates' contributions

I understand that this addendum may be used to adjust my individual grade relative to the team grade

I take full responsibility for my work and any errors in the sections I authored

Signature: Nevaeh Marquez Date: 5/1/26